

Orchestrating the Virtual Tumor Board: A Scoping Review of Multi-Agent AI and Collaborative Decision-Making in Clinical Oncology

Abstract

The rapid evolution of precision oncology has drastically increased the complexity of clinical decision-making, necessitating multidisciplinary collaboration to formulate optimal treatment plans. While traditional Multidisciplinary Tumor Boards (MDTs) are considered the gold standard in oncology, they are resource-intensive and geographically constrained. The transition to Virtual Tumor Boards (VTBs) mitigated some of these logistical barriers, paving the way for digital integration. Concurrently, the advent of Artificial Intelligence (AI) and Large Language Models (LLMs) has introduced the potential for automated clinical decision support. However, isolated single-agent AI systems often fall short of replicating the nuanced, multi-stakeholder deliberations characteristic of human tumor boards. This scoping review evaluates the paradigm shift toward Multi-Agent Systems (MAS) in clinical oncology. We analyze the architectural transition from single-agent diagnostic tools to multi-agent generative frameworks that simulate the collaborative dynamics of human MDTs. By reviewing current approaches, fidelity assessments, and recent deployments, this paper highlights the potential of MAS to enhance decision-making precision while outlining critical limitations regarding data bias, accountability, and real-world clinical deployment.

Keywords: Multi-Agent Systems, Virtual Tumor Board, Clinical Decision Support, Artificial Intelligence, Oncology, Large Language Models, Multidisciplinary Team.

1. Introduction

Modern clinical oncology is characterized by a rapid influx of novel therapeutics, genomic sequencing data, and shifting clinical guidelines. The sheer volume of information has surpassed the cognitive processing capacity of individual clinicians, solidifying the Multidisciplinary Tumor Board (MDT) as an essential component of modern cancer care. MDTs integrate the expertise of medical oncologists, radiation oncologists, surgical oncologists, pathologists, and radiologists to synthesize holistic patient management strategies.

Despite their clinical efficacy, traditional MDTs face significant scalability challenges. They are constrained by scheduling conflicts, geographic silos, and a high administrative burden. To address these bottlenecks, healthcare institutions increasingly adopted Virtual Tumor Boards (VTBs). VTBs democratize access to specialist care, particularly in resource-limited settings and complex subspecialties. However, while VTBs solve spatial and temporal barriers, they do not inherently reduce the cognitive load placed on physicians.

Recently, Artificial Intelligence (AI) and Large Language Models (LLMs) have emerged as powerful tools for clinical decision support. Early implementations utilized single-agent AI models to process electronic health records, query guidelines, and predict outcomes. Yet, real-world oncology is not merely an extraction task; it involves complex reasoning, negotiation of risk, and consensus-building. Consequently, the research frontier has shifted toward Multi-Agent Systems (MAS)—computational frameworks where distinct AI "agents," endowed with specialized knowledge, interact, debate, and collaborate to reach a clinical consensus. This scoping review maps the intersection of VTBs and MAS, analyzing how collaborative AI orchestrates multidisciplinary oncology care.

2. Background, Related Work, and Literature Review

2.1 The Evolution of the Virtual Tumor Board (VTB)

The decentralization of tumor boards via telecommunication platforms initiated the digital transformation of multidisciplinary care. The implementation of VTBs has been shown to directly impact clinical management pathways and improve the adoption of optimized therapies. For example, following the implementation of VTBs, clinical practices demonstrated significant shifts toward more efficient and evidence-based radiotherapy regimens.

The success of VTBs extends across various complex oncology domains. In highly specialized fields such as neuro-oncology, the emergence of VTBs has provided critical opportunities to pool scarce expertise and standardize care for rare malignancies. Similarly, large-scale public health initiatives have leveraged VTBs to bridge the gap between academic tertiary centers and community hospitals. The National Cancer Grid in India utilized VTBs specifically for head and neck cancers, demonstrating that virtual multidisciplinary care could be effectively scaled at a national level to standardize complex oncological decisions.

2.2 The Introduction of Artificial Intelligence in Oncology

As VTBs normalized digital clinical collaboration, AI was introduced to assist in data curation and guideline matching. Early clinical decision support systems relied heavily on deterministic rules or classical machine learning models for outcome prediction. The recent leap in generative AI—specifically Transformer-based LLMs—has enabled machines to process unstructured clinical narratives, pathology reports, and complex medical literature. However, single-agent models frequently exhibit critical vulnerabilities, including "hallucinations" (generating medically inaccurate information), poor multi-step reasoning, and an inability to weigh conflicting clinical priorities.

3. Main Thematic and Technical Discussion: Multi-Agent AI Architectures

3.1 The Limitation of Single-Agent Paradigms

Single-agent AI systems attempt to function as a monolithic "all-knowing" entity. In clinical oncology, this approach is fundamentally flawed. An optimal treatment plan requires balancing aggressive surgical resection against systemic toxicity, prioritizing quality of life, and evaluating long-term survival statistics. A single LLM typically outputs a blended, generalized response that lacks the rigorous, adversarial review process of human experts.

3.2 Principles of Multi-Agent Systems (MAS) in Oncology

Multi-Agent Systems replicate the structural dynamics of a human tumor board. In a MAS architecture, a primary LLM is instantiated into multiple distinct sub-agents, each assigned a specific persona, system prompt, and access to domain-specific tools (e.g., Retrieval-Augmented Generation interfaces linked to discrete databases).

A standard MAS Virtual Tumor Board orchestration typically includes:

1. **The Extraction Agent:** Responsible for structuring unstructured Electronic Health Record (EHR) data, lab results, and imaging reports into a standardized clinical vignette.
2. **Domain Specialist Agents (e.g., Medical Oncology Agent, Surgical Agent, Radiology Agent):** These agents independently analyze the vignette using their specific knowledge bases. For instance, the Surgical Agent queries surgical operability criteria, while the Medical Oncology Agent searches the latest genomic-targeted therapies.
3. **The Moderator/Coordinator Agent:** Similar to a lead clinician, this agent synthesizes the recommendations, identifies discrepancies between specialist agents, and forces iterative debate until a consensus is reached.
4. **The Safety/Review Agent:** An independent agent that strictly cross-references the final consensus against current guidelines (e.g., NCCN, ASCO, ESMO) to catch hallucinatory or contraindicated recommendations.

By compartmentalizing roles, MAS reduces the cognitive burden on any single algorithmic processing step and dramatically lowers the risk of hallucinations through cross-agent validation.

4. Current Approaches and Developments

4.1 Assessing AI Fidelity to Human Decision-Making

A critical metric for evaluating MAS in oncology is its concordance (fidelity) with retrospective human MDT decisions. Recent studies have rigorously tested AI against human consensus. In gastric cancer management, researchers evaluated the fidelity of AI to actual multidisciplinary tumor board recommendations, demonstrating varying degrees of alignment depending on case complexity and the clarity of existing clinical guidelines.

Similar observational studies have been conducted in broader surgical oncology contexts. Retrospective analyses assessing the concordance between AI-generated recommendations and human multidisciplinary tumor boards in general surgery indicate that while AI overlaps significantly with human decisions in standard cases, it frequently diverges during critical decision points involving borderline resectability and complex staging. These findings underscore that AI functions best as a collaborative support layer rather than an autonomous replacement.

4.2 Advancements in Multi-Agent Clinical Platforms

The literature highlights the rapid development of specialized MAS frameworks tailored to oncology. Innovations include self-evolving systems designed explicitly for structured clinical decision-making in multi-cancer environments, showcasing the ability of collaborative agents to iteratively improve their reasoning pathways.

Furthermore, comprehensive reviews analyzing the role of AI within multidisciplinary tumor boards indicate that AI tools significantly enhance information synthesis, guideline alignment, and workflow efficiency. Across various cancers, task-focused AI tools have been shown to support MDT preparation and deliberation, directly enhancing decision-making and clinical outcomes in oncology settings.

5. Research Challenges and Limitations

Despite the architectural advantages of Multi-Agent Systems, several critical limitations restrict their immediate clinical deployment:

1. **Algorithmic Bias and Generalizability:** AI agents are heavily reliant on their underlying training corpora. If the guidelines, clinical trial data, and historical EHRs used for training predominantly feature specific demographic cohorts, the resulting MAS recommendations may inadvertently degrade care for underrepresented populations.
2. **The "Black Box" of Multi-Agent Negotiation:** While human MDTs document the rationale behind their final decisions, capturing the emergent logic generated by multiple AI agents debating in latent space is difficult. Explainability remains a primary bottleneck; clinicians cannot legally or ethically accept a recommendation if the synthesis pathway is opaque.
3. **Limitations in Nuance and Subjectivity:** AI currently struggles to quantify subjective patient factors, such as frailty, socioeconomic barriers to treatment adherence, and explicit patient preferences regarding end-of-life care. As noted in surgical oncology studies, AI often leans toward rigid guideline adherence, whereas human boards adeptly exclude aggressive surgical options based on nuanced clinical intuition.
4. **Regulatory and Legal Liability:** The integration of autonomous debating agents creates a fragmented liability landscape. If a Multi-Agent system reaches a consensus that results in an adverse patient outcome, attributing medical malpractice—whether to the software developer, the underlying LLM provider, or the clinician who accepted the output—remains an unresolved legal challenge.

6. Research Gaps and Future Directions

The current literature predominantly focuses on retrospective concordance studies. To advance the field, future research must pivot toward:

- **Prospective, Randomized Clinical Trials:** There is a glaring lack of prospective trials evaluating the real-world impact of MAS-assisted Virtual Tumor Boards on patient survival (Overall Survival and Progression-Free Survival) and institutional cost metrics.
- **Multimodal Agentic Integration:** Current MAS models rely heavily on text-based LLMs. Future virtual tumor boards must incorporate multimodal agents capable of

directly processing and interpreting raw histological slides and radiological DICOM images, rather than relying on human-generated text summaries of those images.

- **Dynamic, Federated Knowledge Updating:** Oncology guidelines evolve rapidly. Research is needed to develop federated multi-agent systems that securely update their knowledge bases via real-time global clinical trial data streams without requiring constant foundational model retraining.
- **Human-AI Teaming Frameworks:** The most effective future paradigm is not autonomous AI, but a symbiotic human-AI tumor board. Designing user interfaces that allow human clinicians to seamlessly interject, steer, or override AI-agent deliberations is a critical next step.

7. Conclusion

The orchestration of the Virtual Tumor Board using Multi-Agent Artificial Intelligence represents a profound structural shift in clinical oncology. By transitioning away from single, error-prone language models to specialized, collaborative agentic frameworks, the medical community can simulate the rigor, debate, and consensus-building of a human multidisciplinary team. Current literature confirms that VTB platforms successfully scale expertise geographically, while AI-driven decision systems are beginning to demonstrate high concordance with human experts in specific malignancies.

However, AI-generated recommendations still struggle with the nuanced realities of borderline cases and subjective patient factors. Ultimately, Multi-Agent Systems are not poised to replace the human oncologist. Instead, they serve as an invaluable cognitive exoskeleton—rapidly synthesizing literature, ensuring guideline adherence, and preparing comprehensive data so that human clinicians can focus entirely on complex, empathetic, and individualized patient care.

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